IE643 Lecture 7

Announcement:

Quiz-1

Date: 1st September 2026 (Tuesday)

Venue: In class.

Syllabus: Contents till Perceptron mistake bound analysis.

Open scribes/notes. More instructions to be posted in moodle.

Recap:

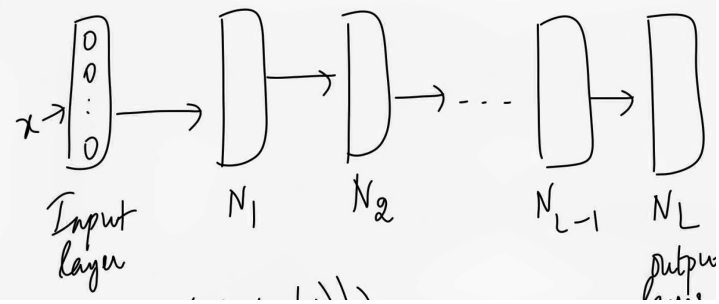


Figure 1. A chain of $N_1 \to N_2 \to \cdots \to N_{L-1} \to N_L$ boxes, labeled "Input layer" (input vector $x \in \mathbb{R}^{n_d}$) on the left, "Output layer" on the right.

$$\hat{y} = a^L = \phi(W^L; \phi(W^3\phi(W^2\phi(W^1 x))))$$

$e(y, \hat{y}) \to$ loss fn to compare true and predicted labels

Supervised learning

$$D = \{(x^i, y^i)\}_{i=1}^N$$

$$x \ni x^i \in \mathbb{R}^d$$
$$y \ni y^i \in \{+1, -1\}$$

$$h : X \to Y$$

MLP weights

train a multilayered perceptron to find h_{MLP}

$$\min E = \sum_{i=1}^{N} e(y^i, \hat{y}^i)$$

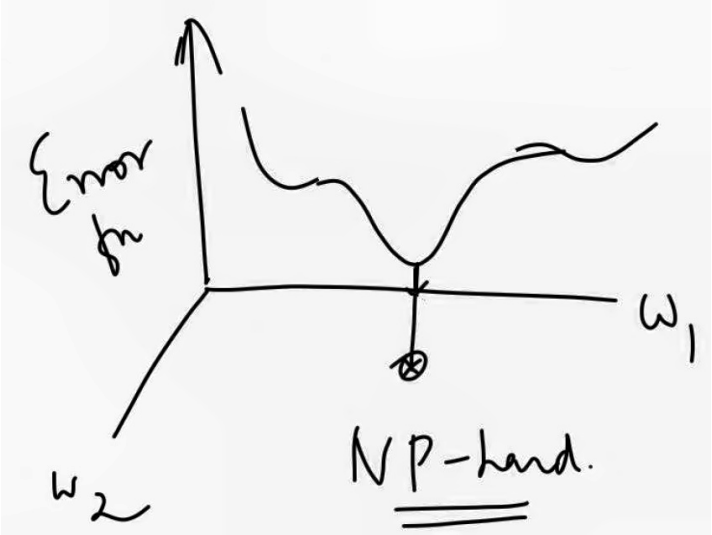


Figure 2. A wavy error curve with a local minimum marked at (w_1, w_2), labeled "NP-hard" and "local minimum".

minimization is over **weights of MLP**

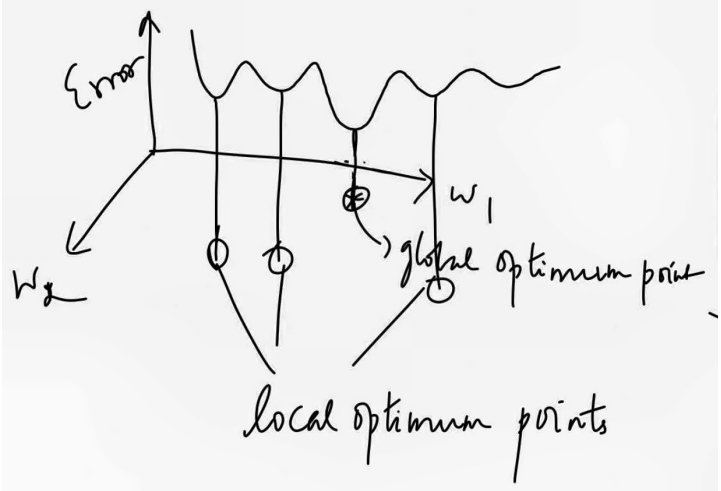


Figure 3. A wavy error curve with local minima marked at w_1, w_2, etc., labeled "local optimum points" — one of these is the global optimum point.

Iterative procedure to find a local minimum point

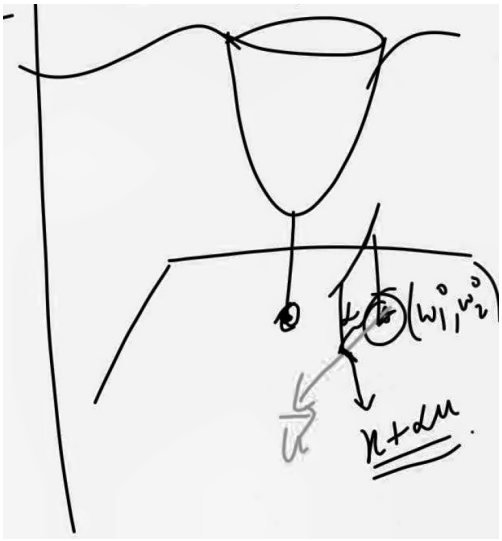


Figure 4. A funnel-shaped bowl (loss surface) with a ball rolling down from w to $w + \Delta w$.

Directional derivative:

For a fn. $f : \mathbb{R}^d \to \mathbb{R}$, the directional derivative along some **non-zero** direction $u \in \mathbb{R}^d$ at a point x is defined as

$$f'(x;u) = \lim_{\alpha \downarrow 0} \frac{f(x+\alpha u) - f(x)}{\alpha}$$

Gradient

Gradient of a fn at a point x

$$\nabla f(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_d} \end{bmatrix}$$

$$f'(x;u) = \langle \nabla f(x), u \rangle$$

$$u = -\nabla f(x) = -\|\nabla f(x)\|^2$$

If $\nabla f(x) \neq$ zero vector:

$$f'(x;u) = \langle \nabla f(x), -\nabla f(x) \rangle$$

Descent Direction

$$f'(x; -\nabla f(x)) < 0$$

Descent direction:

A non-zero direction $u \in \mathbb{R}^d$ is called a descent direction of f at x if:

$$f'(x; u) < 0$$

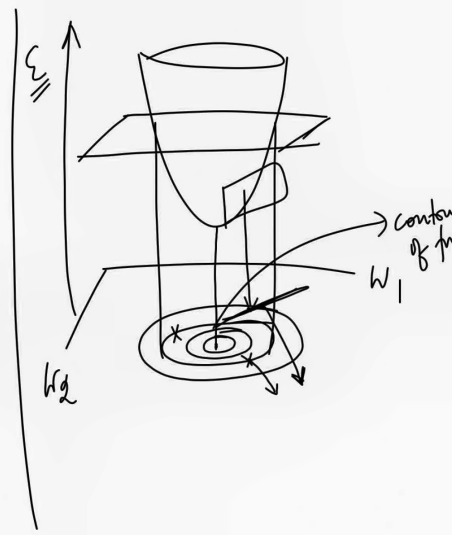


Figure 5. A 3D surface with level curves (contours of f) labeled w_1, w_2, and an arrow pointing downward into the minimum, illustrating a descent direction.

Negative gradient direction & step-length/learning rate

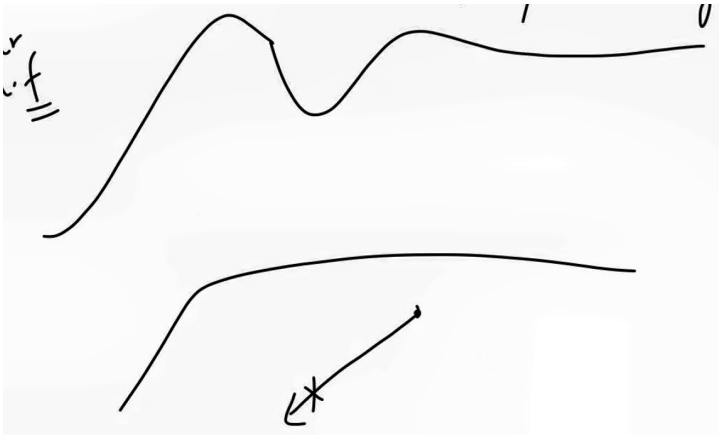


Figure 6. A wavy curve with an arrow pointing left from a marked point, labeled "step length/learning rate."

$$f'(x, u) = \lim_{\alpha \downarrow 0} \frac{f(x + \alpha u) - f(x)}{\alpha}$$

Property:

If u (a non-zero vector) is a descent direction of fn $f : \mathbb{R}^d \to \mathbb{R}$ at x, then:

there exists some $\epsilon \geq 0$ such that for every $\alpha \in [0, \epsilon]$, $\lim_{\alpha \downarrow 0} \frac{f(x+\alpha u)-f(x)}{\alpha} < 0$

$$f(x + \alpha u) < f(x) \quad \Rightarrow \quad f'(x; u) < 0$$

Gradient Descent

Gradient descent to solve

$$\min_{x \in \mathbb{R}^d} f(x) \qquad f : \mathbb{R}^d \to \mathbb{R}$$

$\to$ Start at $x^0 \in \mathbb{R}^d$

$\to$ for $k = 0, 1, 2, \ldots$

- find a descent direction u^k
- find step length α_k
- $x^{k+1} \leftarrow x^k + \alpha_k u^k$

1. Local minimum 2. Saddlepoints

$\to$ **-ve gradient** $\to$??

|| 85p' ||

$\min_w E = \sum_{i=1}^{N} e(y^i, \hat{y}^i)$ $\quad \nabla E = \sum_{i=1}^{N} \nabla e(y^i, \hat{y}^i) \to$ full gradient, $\nabla E = \sum_{i=1}^{N'} \nabla e(y^i, \hat{y}^i) \to$ sample-based gradient

for $k = 0, 1, 2, \ldots$ - find descent dirn using $-\nabla E(w^k)$ - find a suitable step length α_k - $w^{k+1} \leftarrow w^k + \alpha_k(-\nabla E(w^k))$

$\to$ zero-gradient check to stop this process ($\nabla E(w^{k+1}) \approx 0$)

$$10^{-5}, 10^{-4}, \ldots$$

ϵ_k: fixed to be small value close to 0, typically 10^{-8}.

$\to$ step-length finding procedures

Stochastic gradient descent

parameter regime: 10^6, 10^8, ...

$$D = \{(x^i, y^i)\}_{i=1}^N$$

$\rightarrow w^0$

$\rightarrow$ for $k = 0, 1, 2, ...$

$\hookrightarrow$ Compute a **stochastic gradient**

- Randomly choose $j \in \{1, 2, 3, ... N\}$
- Compute $\nabla e(y^j, \hat{y}^j)$

$$w^{k+1} \leftarrow w^k + n_k(-\nabla e(y^j, \hat{y}^j))$$

Stochastic approx. $\Rightarrow$ convergence to the local opt/saddle pt.

might not lead to descent

Mini-batch SGD

$$D = \{(x^i, y^i)\}_{i=1}^N$$

$\rightarrow w^0$

$\rightarrow$ for $k = 0, 1, 2, ...$

$\hookrightarrow$ Sample mini-batch of samples:

$$B_k \subseteq \{1, 2, ..., N\}$$

for every $j \in B_k$: compute $\nabla e(y^j, \hat{y}^j)$

$\hookrightarrow w^{k+1} \leftarrow w^k + \eta_k \left(-\sum_{j \in B_k} \nabla e(y^j, \hat{y}^j)\right)$

$\hookrightarrow$ convergence in asymptotic sense

Adam

SGD with momentum

$$m_k = \alpha \cdot \text{grad}_k + (1 - \alpha) m_{k-1}$$

generalize perf

$$u^k = \alpha(\text{mini-batch gradient}) + (1 - \alpha) \cdot \text{past gradients}$$

Adam:

$$w^{k+1} \leftarrow w^k + \eta_k \left(-\frac{\text{running avg of gradients}}{\sqrt{\text{running avg of sq. comp of gradients}}} + \epsilon \right)$$

AMSgrad

AMSGrad

$$m_k = \beta_1 \cdot \text{grad}_k + (1 - \beta_1) m_{k-1}$$
$$v_k = \beta_2 (\text{grad}_k^2) + (1 - \beta) v_{k-1}$$

$$w^{k+1} \leftarrow w^k + \eta_k \left(-\frac{m_k}{\sqrt{v_k} + \epsilon} \right)$$

Sample-based gradient

$$\nabla_l (y^j, \hat{y}^j)$$

Computation of sample-based gradient

$$\hat{y}^j = \phi(W^L \phi(W^{L-1} \phi(\cdots \phi(W^2 \phi(W^1 x^j)) \cdots)))$$

$$D_e(y^j, \hat{y}^j) \rightarrow w_1, w_2, w_3, \ldots, w_L$$

auto-differentiation

Forward Chain Rule

i^{th} neuron in layer l, j^{th} neuron in layer $l-1$: w_{ij}^l

$$\frac{\partial e}{\partial w_{ij}^l} = \frac{\partial e}{\partial z_i^l} \cdot \boxed{\frac{\partial z_i^l}{\partial w_{ij}^l}}$$

forward chain rule:

$$z_i^l = \sum_{k=1}^{N_{l-1}} w_{ik}^l a_k^{l-1}$$
$$a_i^l = \phi(z_i^l)$$
$$= \frac{\partial e}{\partial z_i^l} a_j^{l-1}$$

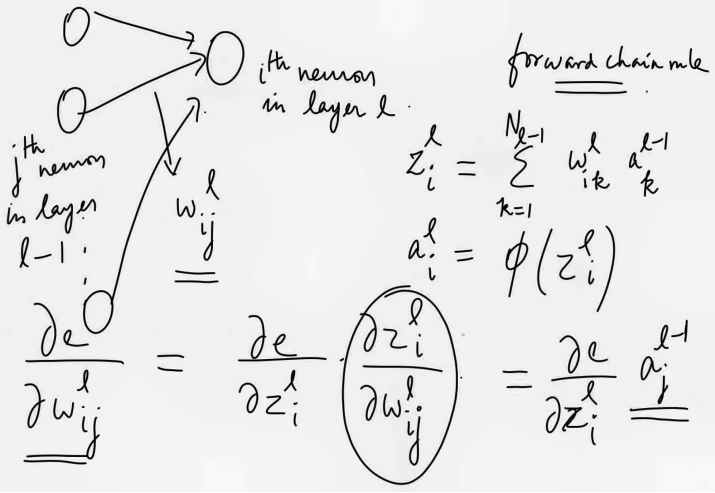


Figure 7. A chain of neurons — input layer neuron → j^{th} neuron in layer $l-1$ → i^{th} neuron in layer l.

$$\frac{\partial e}{\partial z_i^l} = \frac{\partial e}{\partial a_i^l}\left(\frac{\partial a_i^l}{\partial z_i^l}\right) = \frac{\partial e}{\partial a_i^l} \cdot \phi'(z_i^l)$$

$$\frac{\partial e}{\partial a_i^l} = \sum_{m=1}^{N_{l+1}} \frac{\partial e}{\partial z_m^{l+1}} \cdot \left(\frac{\partial z_m^{l+1}}{\partial a_i^l}\right) = \sum_{m=1}^{N_{l+1}} \frac{\partial e}{\partial z_m^{l+1}} \cdot w_{m,i}^{l+1}$$

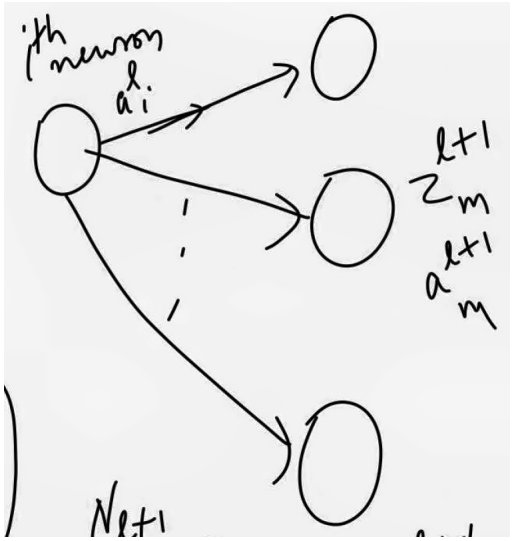


Figure 8. A neuron a_i^l (layer l input) feeding into multiple neurons of layer $l+1$, labeled z_m^{l+1} and a_m^{l+1}.

Chain Rule

$$z_m^{l+1} = \sum_{k=1}^{N_l} w_{mk}^{l+1} a_k^l$$

$$\frac{\partial e}{\partial a_i^l} = \sum_{m=1}^{N_{l+1}} \frac{\partial e}{\partial z_m^{l+1}} \cdot w_{mi}^{l+1}$$

$$= \sum_{m=1}^{N_{l+1}} \frac{\partial e}{\partial a_m^{l+1}} \cdot \frac{\partial a_m^{l+1}}{\partial z_m^{l+1}} \cdot w_{mi}^{l+1}$$

$$= \sum_{m=1}^{N_{l+1}} \frac{\partial e}{\partial a_m^{l+1}} \phi'(z_m^{l+1}) \cdot w_{mi}^{l+1}$$

Auto-differentiation

[Diagram: A hexagonal grid of small circles, each with a dot at its center, arranged in a repeating pattern across the page.]